

1. Scope

This report presents the Day 2 baseline Chidamber & Kemerer (CK) metrics for the eight classes owned by the Desktop GUI and Client Shell work package. These classes collectively implement the non-VR user interface of iDaVIE: file/mask loading, rendering parameter panels, statistics display, source management, paint mode, histogram controls, video recording UI, and tab navigation.

Classes Analysed

Class	File	Role
CanvassDesktop	Assets/Scripts/UI/CanvassDesktop.cs	Orchestrator (O)
DesktopPaintController	Assets/Scripts/UI/DesktopPaintController.cs	Adapter (A)
PaintMenuController	Assets/Scripts/Menu/PaintMenuController.cs	Orchestrator (O)
VideoUiManager	Assets/Scripts/VideoMaker/VideoUiManager.cs	Adapter (A)
HistogramMenuController	Assets/Scripts/Menu/HistogramMenuController.cs	Adapter (A)
HistogramHelper	Assets/Scripts/Menu/HistogramHelper.cs	Domain helper (D)
SourceRow	Assets/Scripts/Menu/SourceRow.cs	Domain helper (D)
TabsManager	Assets/Scripts/Menu/TabsManager.cs	Domain helper (D)

2. CK Metrics Summary

Thresholds from the assignment specification (Section 7.1):

- **WMC** ≤ 20 (domain) / ≤ 40 (orchestrator/adaptor)
- **DIT** ≤ 4
- **NOC** ≤ 5
- **CBO** ≤ 14 (domain) / ≤ 25 (orchestrator)
- **RFC** ≤ 50
- **LCOM (Henderson-Sellers)** ≤ 0.50

Class	Role	WMC	DIT	NOC	CBO	RFC	LCOM_HS
CanvassDesktop	O	63 	1 	0 	47 	118 	0.955 
DesktopPaintController	A	57 	1 	0 	21 	99 	0.940 
PaintMenuController	O	24 	1 	0 	9 	56 	0.919 
VideoUiManager	A	17 	1 	0 	17 	64 	0.863 
HistogramMenuController	A	13 	1 	0 	12 	36 	0.812 
HistogramHelper	D	3 	1 	0 	13 	23 	0.667 
SourceRow	D	3 	1 	0 	3 	11 	0.667 
TabsManager	D	3 	1 	0 	4 	7 	0.467 

 = violation ·  = at/near threshold ·  = within threshold

3. Violation Analysis

3.1 DIT / NOC — No violations

All eight classes sit one level above **MonoBehaviour** (DIT = 1). None have subclasses (NOC = 0). Inheritance is not a concern in this subsystem.

3.2 WMC — 2 violations

CanvassDesktop (63) and **DesktopPaintController (57)** both exceed the orchestrator/adaptor ceiling of 40. These are god classes that absorb file-loading, rendering control, statistics display, paint mode, source management, and VR/desktop bridging into a single **MonoBehaviour**.

3.3 CBO — 1 critical violation

CanvassDesktop (47) nearly doubles the orchestrator threshold of 25. It directly couples to:

- 23 project classes (e.g. `VolumeDataSetRenderer`, `FitsReader`, `DataAnalysis`, `FeatureMapping`)
- 13 Unity/TMPPro UI types
- 7 System library types (`IntPtr`, `StringBuilder`, `Marshal`, etc.)
- 4 Valve.VR types (`SteamVR`, `OpenVR`)

Any change to `FitsReader`, `VolumeCommandController`, `DataAnalysis`, `FeatureMapping`, or `MenuBarBehaviour` ripples directly into this class.

HistogramHelper (13) is borderline against the domain threshold of 14, driven by its OxyPlot charting dependency.

3.4 RFC — 5 violations

Class	RFC	Threshold	Over by
CanvassDesktop	~118	50	+68
DesktopPaintController	~99	50	+49
VideoUiManager	64	50	+14
PaintMenuController	56	50	+6

High RFC correlates directly with high CBO and WMC — these classes call out to too many external types because they own too many responsibilities.

3.5 LCOM — 7 of 8 classes violate (≤ 0.50)

TabsManager (0.467) is the only clean class.

Critical cases:

- **CanvassDesktop (0.955)**: 63 methods, 67 fields, but total field–method access count is only 189. Methods operate on small, largely disjoint subsets of the field set — the textbook signature of a class that should be split. Three fields (`_restFrequency`, `inPaintMode`, `_tabsManager`) are declared but never accessed by any method.
- **DesktopPaintController (0.940)**: 57 methods, 66 fields, access count 226. Same disjoint-access pattern. The `axis` field is touched by 20 methods while `firstEnable`, `colormapHeight`, and `minZoom` are dead.
- **PaintMenuController (0.919)**: Deceptively bad. `_volumeInputController` is accessed by 16 of 24 methods, but 5 fields (`cropstatus`, `featureStatus`,

`oldSaveText`, `paintMenu`, `savePopup`) are declared and never touched — dead weight inflating LCOM.

- **VideoUiManager (0.863)**: `_isPaused` is declared but never accessed.

Low-priority cases:

- **HistogramHelper (0.667)** and **SourceRow (0.667)**: Trivially small classes. Their LCOM violations are artefacts of empty Unity lifecycle stubs (`Start/Update` with no bodies) and publicly-set data fields that the class never reads internally. These inflate LCOM mathematically but do not indicate a design problem.

4. Dead Code Inventory

Class	Dead Fields
CanvassDesktop	<code>_restFrequency</code> , <code>inPaintMode</code> , <code>_tabsManager</code>
DesktopPaintController	<code>firstEnable</code> , <code>colormapHeight</code> , <code>minZoom</code>
PaintMenuController	<code>cropstatus</code> , <code>featureStatus</code> , <code>oldSaveText</code> , <code>paintMenu</code> , <code>savePopup</code>
VideoUiManager	<code>_isPaused</code>
HistogramMenuController	<code>editMinScale</code> , <code>editMaxScale</code>

12 dead fields across 5 classes. These should be removed in the refactoring proposal.

5. Key Coupling Dependencies (CanvassDesktop)

CanvassDesktop's 47-type coupling breaks down as:

Category	Count	Examples
Project types	23	<code>VolumeDataSetRenderer</code> , <code>VolumeInputController</code> , <code>FitsReader</code> , <code>DataAnalysis</code> , <code>FeatureSetManager</code> , <code>FeatureMapping</code> , <code>Config</code> , <code>ColorMapUtils</code>
Unity / TMP	13	<code>TMP_Dropdown</code> , <code>TMP_InputField</code> , <code>Toggle</code> , <code>Slider</code> , <code>Button</code> , <code>Coroutine</code> , <code>PlayerPrefs</code>

System	7	<code>IntPtr</code> , <code>StringBuilder</code> , <code>FileInfo</code> , <code>Marshal</code> , <code>CultureInfo</code>
Third-party	4	<code>StandaloneFileBrowser</code> , <code>ExtensionFilter</code> , <code>SteamVR</code> , <code>OpenVR</code>

6. Implications for Refactoring

The two primary refactoring targets are **CanvassDesktop** and **DesktopPaintController**. Both are god classes with extreme WMC, RFC, and LCOM violations.

The assignment specification (Section 6.6) directs the following splits:

1. **CanvassDesktop** → **MVVM decomposition**: View (Unity 6 UI Toolkit), ViewModel (pure C#), Service Gateway (server communication). Menu structure, panel state, file dialogs, and configuration become separate, composable responsibilities.
2. **File tab**: From direct native-plugin call to ViewModel command via service gateway (Worked Example 1).
3. **Debug tab**: From inline GUI logic to Observer of a structured logging stream (Worked Example 2).

The projected Day 13 metrics for these refactored classes should bring WMC, CBO, RFC, and LCOM within their respective thresholds.

Report prepared by Sub-team 6 Quality Champion · Sprint 1 · iDaVIE Refactoring Assessment 2026